Efficient Connectivity Through Shared Network Infrastructure

Two Concepts for 4G and 5G Cooperation and LEO Satellite Backhaul

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Abstract

This paper presents two distinct concepts for improving connectivity through infrastructure reuse. The first proposes congestion-aware cooperation between 4G and 5G networks to improve utilisation of available radio and backhaul capacity. The second proposes a regional low Earth orbit satellite network using shared ground terminals to serve communities, institutions, businesses and local access networks. Both concepts support internet terminals and connected devices as well as mobile services. Each can be developed independently, with optional coordination where deployed together. The paper defines their problems, proposed solutions, technical boundaries and validation requirements. Performance improvements, coverage and cost advantages remain hypotheses requiring evaluation.

Keywords: 4G, 5G, LTE, NR, LEO satellites, satellite backhaul, internet access, shared infrastructure, network optimisation.

Concept One

Congestion Aware Cooperation Between 4G and 5G Networks

1. Problem

Existing 4G and 5G networks may experience uneven demand. A congested 5G cell can coexist with spare capacity in a nearby 4G cell, yet users may remain on the congested connection because of device capabilities, network configuration or allocation policies.

Expanding radio capacity alone may not resolve the problem when transport links or backhaul are also congested. The engineering challenge is to use available resources more effectively while maintaining service quality.

2. Proposed Solution

Introduce a software coordination layer that evaluates network conditions and selects suitable access configurations for eligible connections.

The controller considers:

- Available radio and backhaul capacity.
- Signal quality and device capabilities.
- Required throughput, latency and reliability.
- Existing user commitments and fairness.
- Energy consumption and switching overhead.

Depending on supported capabilities, it selects native 4G, native 5G or dual connectivity.

Conceptual control flow

> Network and device measurements → Coordination layer → Supported 4G, 5G or dual-connectivity configuration

The coordination layer guides allocation policies, while existing radio schedulers continue to manage radio resources.

3. Operating Principle

A delay-tolerant transfer could use available 4G capacity when 5G is congested. A real-time session could remain on the eligible connection with the lowest measured delay. A high-throughput session could use supported dual connectivity when both networks have suitable resources.

Decisions follow service requirements and measured conditions rather than packet size or network generation alone. Switching thresholds and minimum dwell times prevent excessive movement between connections. Admission controls protect existing users from overload.

This approach applies to compatible smartphones, cellular routers, fixed wireless internet terminals and other cellular devices. Computers and local networks can benefit through a compatible router or access terminal.

4. Technical Boundary

LTE–NR multi-connectivity is already standardised [1]. LTE is the radio technology commonly associated with 4G; NR is the 5G radio interface. The proposed research concerns improved coordination and reuse of compatible infrastructure.

Software cannot be assumed to make arbitrary LTE-only radios decode native NR transmissions. Frequency support, radio hardware, baseband processing, network interfaces and device capabilities remain constraints.

The initial implementation should therefore use supported traffic steering and dual-connectivity mechanisms. Any deeper radio compatibility enhancement requires separate, equipment-specific validation.

5. Expected Benefits and Validation

The hypothesis is that coordinated allocation can improve service consistency and infrastructure utilisation under uneven load.

Evaluate the controller against a tuned existing network using matched traffic, spectrum, devices and service requirements. Measure:

- Delivered throughput and application completion time.
- Tail latency, packet loss and service interruptions.
- Fairness between users.
- Switching overhead and energy per delivered gigabyte.
- Integration and operating costs.

Benefits depend on spare capacity and compatible equipment. The controller cannot create additional spectrum or overcome a saturated shared backhaul link.

Concept Two

Shared LEO Satellite Backhaul for Regional Internet Connectivity

6. Problem

Remote settlements, islands and difficult terrain can make terrestrial backhaul costly or slow to deploy. Communities may have local access infrastructure but lack a reliable connection to wider networks.

Installing a separate satellite terminal for every user can duplicate equipment, power requirements and maintenance. Where demand is geographically concentrated, shared satellite infrastructure offers an alternative.

7. Proposed Solution

Deploy a LEO satellite network that connects shared regional or community ground terminals. Each terminal aggregates demand from nearby settlements, institutions, businesses or access networks.

Connectivity is then distributed through fibre, Ethernet, microwave, Wi-Fi or cellular infrastructure.

Conceptual service path

> Core network and internet ↔ Feeder gateway ↔ LEO satellite network ↔ Shared ground terminal ↔ Local distribution ↔ Internet terminals and connected devices

Feeder gateways connect the space network to terrestrial infrastructure. Shared ground terminals provide satellite connectivity to local service areas.

The architecture supports internet terminals, routers, computers, public access points, connected equipment and mobile devices. Compatible sites may connect through dedicated satellite terminals where sharing is impractical.

Satellite integration and backhaul are established architectural approaches [2]. The proposed focus is joint optimisation of regional satellite capacity, shared ground infrastructure and local distribution.

8. Space and Ground Design

Constellation size is determined by service requirements rather than a fixed satellite count.

The design jointly evaluates:

- Orbital altitude, inclination, planes and satellite count.
- Payload throughput and beam allocation.
- Ground-terminal placement and feeder connectivity.
- Local distribution capacity and peak demand.
- Handover, redundancy and replacement requirements.

Higher payload capacity cannot compensate for a satellite being out of view. Continuous service requires both sufficient visibility and a usable end-to-end network path.

Ground sites should be selected using demand concentration, sky visibility, terrain, power availability, weather exposure and maintenance access. Sharing equipment can reduce duplication, but it can also concentrate failure risk. Critical service areas require backup connectivity or diverse paths.

9. Regional Priority and Wider Use

A deployment could prioritise India or another defined region. Capacity available elsewhere could serve additional ground terminals where coverage, feeder connectivity and service authorisations permit.

Committed regional capacity must remain protected. A regional constellation does not automatically provide continuous global coverage.

Optional inter-satellite links may extend routing choices, but their benefits must justify additional power, mass, complexity and operating cost.

10. Expected Benefits and Validation

The hypothesis is that shared ground terminals and existing local distribution can extend internet connectivity economically in selected underserved areas.

Evaluation should include:

- Orbital visibility at every target site.
- End-to-end capacity during peak demand.
- Outage duration and handover interruptions.
- Weather degradation and equipment failures.
- Local distribution bottlenecks.
- Lifecycle cost and energy per delivered gigabyte.

Cost comparisons should include spacecraft, launches, replenishment, feeder gateways, ground terminals, local distribution, operations and maintenance.

A design is viable only if it meets specified service requirements at a competitive lifecycle cost. A smaller fleet or fewer terminals alone does not establish efficiency.

11. Independent Deployment and Optional Coordination

The concepts address different layers of the connectivity problem.

Concept One improves allocation across compatible terrestrial access networks. Concept Two supplies backhaul to locations that need additional connectivity. Either can operate without the other.

Where deployed together, satellite and terrestrial backhaul measurements can inform the 4G and 5G controller. During congestion, handover or link degradation, the controller can adjust admission and traffic allocation within service limits.

Alternative routes should be used where available. Otherwise, demand must be managed within actual capacity. Management interfaces require authentication, access control and stable fallback behaviour.

12. Research Contribution and Limitations

The proposed contribution is a framework for evaluating infrastructure reuse at two levels: terrestrial resource coordination and shared regional satellite backhaul.

The underlying technologies already exist. The research questions concern implementation efficiency, service quality, resilience and lifecycle cost.

This paper reports no experimental results, guaranteed coverage, quantified savings or established novelty. Numerical acceptance thresholds should be defined before testing, and assumptions and configurations should be documented for reproducibility.

13. Conclusion

Congestion-aware 4G and 5G cooperation offers a testable approach to using existing terrestrial capacity more effectively. Shared LEO backhaul offers a separate approach to extending connectivity to internet terminals, local networks and connected devices in underserved regions.

Both concepts should advance through simulation, controlled prototypes and measured comparisons. Deployment decisions should follow demonstrated service quality, resilience and lifecycle efficiency.

References

[1] 3GPP TS 37.340, version 18.7.0, Release 18. E-UTRA and NR; Multi-connectivity; Overall Description; Stage-2. ETSI TS 137 340, April 2026. Technical specification.

[2] ETSI TR 103 611, version 1.1.1. Satellite Earth Stations and Systems; Seamless integration of satellite and/or HAPS systems into 5G and related architecture options. June 2020. Technical report.